A321NY ATLANTIC (EUROPE) REFERENCE GUIDE

FLIGHT PLANNING

- ❑ Flight PlanFlt/Date STA
- ❑ Flight PlanDesignate
- ❑ TPS ....CCI or WBD Flt/Date/STA
- ❑ Chart Change Notices/NOTAMS
- ❑ Crew List.................................CCI or NS/Flt#/Date/STA
- ❑ AIREP Form
- ❑ Obtain Weather ChartsWSI

(Wind Prog FL 340, High Level Sig Prog, Volcanic Ash)

- ❑ Track MessageEvaluate and crosscheck with flight plan
- ❑ Preflight EFB
- ❑ Current Crew Passports

A321XLR — ETOPS-180:

Max diversion time 180 min (~1,263 nm from suitable airport). Confirm ETOPS authorization in ops specs. ETOPS pre-departure check required per AML.

Track Message Evaluation (NAT)

Check Date/Time (30°W) for validity: EB valid 0100–0800Z, WB valid 1130–1900Z.

Verify TMI and number of parts including amendments. Determine if R Lat SM track. Compare waypoints to flight plan.

PRE-DEPARTURE AT AIRCRAFT

- ❑ Fit for Duty (CCI or ACARS>PREFLT>page 2)
- ❑ Gen Dec and Customs Forms (if required) FA1/Purser Briefing
- ❑ ETOPS Pre-Departure Check (PDC) in AML (also Transit Check, MEL(s)/CDL(s))
- ❑ Water/LAV Service ReportVerify accomplished
- ❑ Security InspectionVerify accomplished
- ❑ ATC Route Clearance Verification
- ❑ CCI – Verify all required CCI documents on iPad
- ❑ Takeoff Briefing including TA, Noise Abatement, Terrain, Engine Out SIDs

SATCOM Setup:

SATCOM #1 → IOC   SATCOM #2 → Controlling OCA

Departure Clearance and Slot Time (Europe):

A slot time defines a 15-minute takeoff window. If ready early, advise ground: “[callsign] is fully ready and can accept an earlier slot, please advise CFMU.” All five conditions must be met before calling: pax/cargo aboard, fueling complete, catering removed, all doors closed, crew ready for pushback.

ACARS ATC clearance (Europe DCL): CDG, DUB, FRA, LGW, LHR, MAN, MUC

- ❑ TSA PA…US Bound Flights…TSA PA:

“Ladies and gentlemen, we request that you comply with the following Transportation Security Administration security directive. During our flight today we ask you to please use the lavatory in your ticketed cabin. Exceptions to this rule apply to those with special needs or medical conditions, elderly persons, or parents with small children. We also ask you not to congregate in groups in any area of the cabin especially around the lavatories. Please contact your flight attendant should you require assistance or have a question about this policy. Thank you for your cooperation.”

CLIMB - CRUISE

- ❑ Update winds when new updates available (0000Z, 0600Z, 1200Z, 1800Z)

NOTE: A321XLR wind update cycle differs from 737 (was 0500/1100/1700/2300Z). Confirm cruise wind upload.

- ❑ Perform Navigation Accuracy Check
- ❑ Review Jeppesen country pages

Position Reports (if ADS is Unavailable)

Use AIREP Form as a reference for the phraseology and order for making position reports.

- ❑ Report each compulsory fix (filled-in solid waypoints only): aircraft ID/position/time/flight level/next compulsory fix/ETA next compulsory fix

If ▲ symbol located at fix: meteorological report required (temperature, wind, any significant weather)

RVSM Altimeter Crosscheck (A321XLR addition)

When established in cruise at or above FL290: crosscheck each PFD altimeter and the standby altimeter. Record results. Repeat hourly. Select AP1 / Transponder 1 for RVSM.

Diversion Airports

- ❑ Continually evaluate weather for enroute diversion airports and arrival routings

Approximately one hour prior to OEP: captain reviews all potential enroute diversion airports using most recent weather.

Weather Radar

- ❑ Ensure weather radar ON during IMC or night

CLIMB-CRUISE – AFTER RECEIPT OF OCEANIC CLEARANCE

Fly the CLEARANCE not the Flight Plan.

❑ PM compares Oceanic Clearance to FMS; notes any differences

❑ PF compares Oceanic Clearance to FMS; differences corrected immediately while PM confirms accuracy

❑ Circle & Tick — one pilot reads waypoint names from FMS aloud; other checks against Oceanic Clearance and MFP, places a circle (O) next to waypoint on MFP or chart

NOTE: Crosscheck FMS coordinates by comparing oceanic waypoint expanded coordinates against flight plan coordinates.

❑ Update winds in FMS (if rerouted)

❑ Set assigned MACH and cleared cruise altitude in FMS

❑ Plot route and label each waypoint

❑ Evaluate potential enroute diversion airport(s) WX/services

❑ Obtain alternate airports WX, forecast, and NOTAMs

❑ Check A/C Systems Status / ETOPS Significant Systems

❑ ETOPS Inflight Verification Check(s) prior to ETOPS entry (e.g., APU inflight start) — AML entry required. Contact Dispatch if unsuccessful.

❑ Verify ETA to OCA Entry Point (compare FMS and FPR)

❑ Verify GPS Primary

❑ Determine Altitude Capability — WAH for KZWY, LPPO

Step Altitudes (MCDU): F-PLN > vertical revision > STEP ALTS [5R] — predicts time/distance for selected altitude.

❑ Log on CPDLC 10–25 minutes prior to airspace entry

A321XLR MCDU path: MCDU MENU > ATSU [6L] > AOC MENU [2R] > ATS > CONNECTION > NOTIFICATION.

❑ HF/SELCAL check (as required)

❑ Verify next leg (circle & tick)

After Oceanic Entry Point (OEP)

❑ HF/SELCAL check (if not already done)

• One HF SELCAL Inoperative. Place the primary HF frequency in the HF radio with the operating SELCAL. In this configuration continuous monitoring of the HF is not required.

• Both HF SELCALs Inoperative. If unable to obtain a SELCAL check on either HF radio, then a pilot will continuously monitor the primary HF frequency.

❑ Enter nearest Diversion Airport on PM PROG Page

❑ Set left VHF radio to 121.5 and right to 123.45

❑ Transponder to 2000 when 30 min. past radar service

❑ Perform strategic lateral offset procedure (SLOP)

❑ CPDLC Confirm Assigned Route Message – PGE+ or PRINT, * appears then SEND

OCEANIC PROCEDURES

Strategic Lateral Offset Procedure (SLOP)

This procedure is recommended to be used on every oceanic flight and is Standard Operating Procedure throughout the NAT Region.

• Fly 1 NM or 2 NM offset / RIGHT only.

• Decision to OFFSET based on TCAS traffic observation.

• Offset using LNAV / NAV, not heading select.

• Do not advise ATC or request ATC clearance.

• Return to course by oceanic exit point.

Approaching Each Waypoint

1. Both pilots must verify that the subsequent full waypoint Lat/Long, course and distance agree with the flight plan or current ATC clearance.

2. Perform the Circle & Tick procedure by drawing a diagonal line thru the circle beside the waypoint if not previously accomplished to indicate verification was accomplished. ⌀

Waypoint Passage

• Verify autopilot is coupled to LNAV

• Confirm the next waypoint becomes active waypoint.

• Verify autopilot is receiving guidance from the FMS, and that it did not revert to heading or track mode.

• Draw a second diagonal line thru the circle beside the waypoint just passed forming an X indicating passage. ⊗

• Record time and fuel on FPR / MFP.

• Complete AIREP form/ Transmit report (if required).

Position Check

Aircraft position and time must be checked 10 minutes (or 2 Degrees) beyond each waypoint passage to verify the correct track is being flown. (Plotting not required)

Midpoint Check

• Check ETA to the next waypoint. If waypoint crossing time changed by 3 minutes or more from the last estimate given to ATC, a revised estimate should be transmitted using CPDLC ATC free text, or voice, as appropriate.

• Check for satisfactory fuel quantity/balance and trend.

Track Exit

• Confirm ATC Clearance

• Request desired altitude and MACH from ATC.

• Confirm SLOP offset is zero.

• Entering Canadian airspace: use HEAVY on first call to each new center frequency.

• Entering Maastricht EDYY: CPDLC logon mandatory if available

Oceanic Clearance – Eastbound

Note: For all OCAs except Shanwick, if oceanic clearance is not received, continue into oceanic airspace.

5a. Gander ACARS Procedures

Send an ACARS oceanic clearance request 30 to 90 minutes (90 better than 30) prior to entry into oceanic airspace. All ACARS clearances must be accepted. If amendments needed, revert to voice procedures. (Expect clearance within 40 to 55 minutes.)

If data link clearance is not received within 30 minutes of entry point, revert to voice.

5b. Flights Transiting WATRS (NY Oceanic) into NAT MNPS Airspace

1. Clearance received at the departing station (voice or PDC) is a valid clearance through NAT MNPS airspace.
2. No other route clearance will be issued unless there is a change.
3. ATC will assign an altitude and Mach # prior to Oceanic Entry Point.
4. These two items plus the pre-departure clearance routing constitute a valid clearance.

5c. Voice Procedures

All crewmembers in the cockpit should copy and crosscheck any voice clearance.

— Initial contact to clearance delivery: “Gander Clearance, American 123, request track Whiskey, flight level 370, Mach .82, estimating CARPE at 0440, max flight level 390.”

— Oceanic clearance on organized track: “American 123, CARPE, track Whiskey, flight level 370, Mach decimal 82, TMI 182.”

Note: Random route or clearance given over HF, full readback.

Gander OCA

1. Contact Gander Within 200 NM of specified Clearance Delivery location using a frequency listed in the Track Message remarks section.
2. Request clearance, use clearance request format.
3. Read back clearance using readback format. (Acknowledge receipt via voice: “Track, FL, MACH, TMI.”)

New York OCA

1. ATC will normally issue altitude and speed assignment unsolicited.
2. The last assigned route, altitude, and speed are to be maintained.
3. Expect to receive oceanic clearance on the last radar sector frequency within the domestic area.

Be prepared to give WAH

5d. Eastbound Techniques

Entering Gander OCA - CZQX

Be ready to write down HF Primary and Secondary frequencies.

Initial VHF contact when told to contact Gander Radio, example:

- Gander RADIO, <call sign>
- after acknowledgment <call sign> C-P-D-L-C, <OCA Name> NEXT

Entering New York OCA – KZWY

Transitioning from New York to Gander, you may be told to contact Gander at 4430 North on <HF Freq.>. The boundary is at 44° 30’ North, shown on JeppFD-Pro

ENTERING LAST OCA, example:

- <OCA Name> RADIO on <HF Freq>
- after acknowledgment <call sign> C-P-D-L-C, [TRACK __, OR Random Route AND last TWO Exit Points] SELCAL CHECK XX-XX

CHANGE SATCOM #2 to new OCA

Oceanic Clearance – Westbound

6a. Shanwick and Santa Maria ACARS Procedures

CAUTION – DO NOT ENTER SHANWICK OCEANIC AIRSPACE WITHOUT OCEANIC CLEARANCE

Send ACARS oceanic clearance request no earlier than 90 minutes - no later than 30 minutes prior to oceanic entry point utilizing format below. Expect clearance within 15 minutes.

6b. Voice Procedures

All crewmembers in the cockpit should copy and crosscheck any voice clearance.

— Initial contact to clearance delivery: “Shanwick Clearance, American 123, request track Bravo, flight level 370, Mach_.82, estimating BEDEX at 0440, max flight level 390.”

— Oceanic clearance on organized track: “American 123, BEDEX, track Bravo, flight level 370, Mach decimal 82, TMI 182.”

Note: Random route or clearance given over HF, full readback.

6b1. Shanwick OCA – VHF 123.95

1. Contact Shanwick at least 40 minutes before the ETA for the OEP.
2. Request oceanic clearance on VHF.
3. Standby on VHF or HF SELCAL for clearance readback and confirmation.

6b2. Reykjavik OCA – Entry over RATSU - VHF 127.85

1. Do not call or send ACARS request to Shanwick.
2. 20 to 25 min. from FIR boundary call Iceland Radio on 127.85 for oceanic clearance.

6b3. Santa Maria OCA – VHF 132.07

1. Contact Santa Maria < 40 MIN before the ETA for the OEP.

2. Standby on VHF or HF SELCAL for clearance readback and confirmation.

6b4. Bodo OCA

1. Bodo OCA control on VHF 127.725, 10 min. prior to OEP or Bodo radio on NAT D family HF, 30 min. prior to OEP.

6c. WESTBOUND Techniques

Enroute Center will provide HF Frequencies OR will switch you to Radio.

Approaching Track Entry Fix, be ready to write down HF Frequencies.

Entering OCA - Shanwick EGGX - Santa Maria LPPO

Example:

- <OCA Name> RADIO <call sign> on <HF Freq>
- after acknowledgment - <call sign> C-P-D-L-C, <OCA Name> NEXT - SELCAL CHECK XX-XX

If LPPO sends “when can you climb” message, respond via ATC MENU>REPORTS>MSG MODIFY to be able to clear LPPO’s message (MSG MODIFY only available after you receive message)

ENTERING NEXT OCA, Going Thru Multiple OCAs (Ex. New York-Gander-Shanwick)

Use Same Phraseology as Entering OCA Westbound Above

CHANGE SATCOM #2 to new OCA

ENTERING LAST OCA, example:

- <OCA Name> RADIO on <HF Freq>
- after acknowledgment <call sign> C-P-D-L-C, [TRACK __, OR Random Route AND last TWO Exit Points] SELCAL CHECK XX-XX

CPDLC/ADS Formats

1. Log on CPDLC 10 to 25 minutes prior to OEP using procedures in FM Part II, Chapter 2 - Communications.
2. Entering a participating FIR, you will be instructed to contact the FIR's radio facility for HF frequency assignment and to establish a SELCAL watch. On initial contact use the phrase C-P-D-L-C (or ADS if only ADS is available) after the flight call sign. If the flight will exit the

current FIR into oceanic airspace, provide the name of the next FIR. Example: New York ARINC, American XX C-P-D-L-C, Gander next, request SELCAL check CKFM.
3. Exiting the FIR into domestic airspace, provide the flight's track letter (or if on a random route, the last two fixes in the cleared route).

Example: Gander radio, American XX C-P-D-L-C (or A-D-S), Track Bravo, request SELCAL check DMCS. Or, example: Gander radio, American XX C-P-D-L-C (or A-D-S), SCROD, VALIE, request SELCAL check DMCS.

FIR Boundary – CPDLC Log On

EDMONTON................... CZEG
GANDER DOMESTIC ....CDQX
GANDER OCEANIC...........CZQX
LONDON.........................EGTT
MAASTRICT…………………...EDYY
MONCTON ....CZQM
MONTREAL.....................CZUL
NEW YORK.......................KZWY
REYJAVIK.........................BIRD
SANTA MARIA ....LPPO
SCOTTISH FIR .... EGPX
SHANNON........................EISN
SHANWICK.......................EGGX
TORONTO.........................CZYZ
WINNIPEG........................CZWG

Oceanic Clearance via Datalink Techniques

Request Oceanic Clearance as CLOSE TO but NO GREATER than 90 MINUTES prior to entry, but > 30 minutes from oceanic boundary

MCDU Menu - ATSU[6L] - AOC Menu[2R] – Messages [2R] – OCEANIC CLNC Request[4L] (or AOC Menu – ATS – OCEANIC CLX)

(SELECT ATC FIR) select OCEANIC CENTER

On AOC OCEAN CLR REQUEST page, fill in prompts:

- Ensure MCDU is displayed [5R].
- ENTER OPTIONAL FREE TEXT. Remarks may indicate the preferred alternative to the requested clearance and maximum flight level (e.g. Max F380) that can be accepted at the boundary.

The crew should expect A CLEARANCE CONFIRMED message within a few minutes. If no confirmation is received within 15 minutes revert to voice procedures.

EXPECT clearance 1 hr. prior to entry EAST; 10 Min. from request WEST (VOICE, if within 30min of entry point EAST, 15 min from request or entry point, WEST)

ACCEPT the Clearance, THEN request any changes required, Print, then:

- PF and PM separately VERIFY each point expanded coordinates on FMS to Track Msg
- SEND copy of clearance to OCC EXCEPT from Gander
- SET Constant MACH for Track
- FILL IN AIREP Form - another pilot to check

Status Message MUST be received - Clearance Confirmed

If not received contact via voice immediately - <center> American # - Standing By

Notify OCEANIC Center 10-25 minutes prior to ENTRY for CPDLC.

If DCDU does not show ACTIVE you still may have capability. ADS is always active. A SELCAL requesting missing position report will confirm the system is not working.

Voice Clearance Request and Readback Formats

CLEARANCE REQUEST
American XX requesting oceanic clearance, estimating (entry fix) at ____Z, requesting FL___, Mach___.
READBACK – NAT Clearance (abbreviated)
American XX is cleared via Track__, TMI____, FL___, Mach___. If any doubt exists as to the TMI or NAT coordinates, request complete track coordinates from OAC.
READBACK – Random Route or Via Flight Planned Route Clearance
American XX is cleared via, <entry fix>, <fix>, <fix>, <fix>, <fix>, <exit fix>, FL___, Mach___.

ARRIVAL BRIEFING

❑ Review airport specific 10-7 page unique engine failure during missed approach procedures
❑ Review transition altitude, country, and airport specific speed restrictions
❑ TERR feature on for any airport with significant terrain features

Frankfurt Arrival Communications
Switched to Frankfurt Director — STATE ONLY: Frankfurt Director + Call Sign
Switched to Tower — STATE ONLY: Call Sign + Runway

All UK Airports
Tell Director: ATIS + A321. Review Constant Descent Approach (CDA), Distance to Go (DTG), Free Speed.
Taxi lights ON while taxiing at European airports.

International Terminal Operations
Return to gate/remote parking: ability to deplane no later than 4 hours international. Coordinate deplane PA; notify dispatcher via ACARS plain text message.

DEPARTING TRACK - CONTINGENCIES

If immediate action is not required, request an amended ATC clearance. Otherwise:
1. Leave the assigned route by turning at least 45 DEG to the left or right.
a. If Unable to maintain altitude, minimize altitude loss until 15 NM offset, Descend below FL280 before diverting, Fly normal altitude + 500'
b. If Able to maintain altitude, after 10 NM off (on way to 15NM offset), climb/descend+ 500', If intention to acquire opposite direction offset track,
consider turning more than 180 DEG in order to re-intercept the offset contingency track.
2. Accomplish appropriate checklists.
3. 4. Advise other aircraft on 121.5, turn exterior lights on.
Advise ATC, (MAYDAY – Grave & Imminent danger) or (PAN PAN - Urgency) preferably
spoken 3 times, shall be used as appropriate. Distress Freq – VHF 121.5, HF 4125
(Maritime Mobil Service), SATCOM 83642182 (Maritime Services)
FOR ADDITIONAL INFORMATION SEE THE ATLANTIC ORIENTATION CHART

Loss of Engine

To minimize initial descent:
1. Accomplish Unable to Maintain Altitude – Loss of Engine Thrust (QRH).
2. FMC ENG OUT / GREEN DOT speed may be used until 15 NM offset.
3. Turn towards a suitable airport when appropriate (below MNPS (FL280)).
4. Select cruise FL that differs ± 500' from normal..

Loss of Cabin Pressure

CAUTION – See procedures on Greenland Critical Terrain Orientation Chart.

Other Emergencies (Medical, Passenger, etc.)

If Able to maintain altitude, after 10 NM off (on way to 15NM offset), climb/descend+ 500', If Intention to acquire opposite direction offset track, consider turning more than 180 DEG in order to re-intercept the offset contingency track.

WEATHER DEVIATIONS

- Request weather deviation clearance from ATC.
- If clearance is denied or no communication is established and if the deviation is:

A. 10 nm or less: remain at ATC assigned FL

B. Deviation greater than 10 nm:

Track East (000–179°): Left of course – DESCEND 300’ / Right of course – CLIMB 300’

Track West (180–359°): Left of course – CLIMB 300’ / Right of course – DESCEND 300’

Memory device: “The South will rise again” — if turning south, CLIMB.

- Advise ATC and other aircraft on 121.5 and/or 123.45 of intentions. Turn on external lights. Maintain traffic watch visually and via TCAS.
- When returning to route/track, be at assigned FL when within 10 nm of centerline.
- Keep ATC updated on intentions and inform when deviation is complete.

NAV ACCURACY CHECK △ ENTRY EXIT

Station	Primary	Secondary	SelCal
/	/		△
/	/		△
/	/		△

NORTHERN

AIREP FORM

SELCAL: ______________________

Station	Primary	Secondary	SelCal
/	/		△
/	/		△
/	/		△

STATION:								
FREQ / TYPE: (Voice/ADS/CPDLC)			△					
CALLSIGN	Position, Callsign	Position, Callsign	Position, Callsign	Position, Callsign	Position, Callsign	Position, Callsign	Position, Callsign	Position, Callsign
POSITION Lat / Long								
Time:								
FLIGHT LEVEL	Flt Lvl	Flt Lvl	Flt Lvl	Flt Lvl	Flt Lvl	Flt Lvl	Flt Lvl	Flt Lvl
ESTIMATING Lat / Long	Est	Est	Est	Est	Est	Est	Est	Est
Time:								
NEXT Lat / Long	Next	Next	Next	Next	Next	Next	Next	Next
FUEL	Fuel	Fuel	Fuel	Fuel	Fuel	Fuel	Fuel	Fuel

* Temp:								
* Wind:	/	/	/	/	/	/	/	/
** Sig WX (Opt):								

Comm Notes

* ***"Not Required in North Atlantic"***

** ***"Severe Turbulence Reports Required"***